RRR week

Richardson extrapolation

$$N_1(h) = M - K_1 h - K_2 h^2 - \dots$$

- M ↳ what we want to compute
- $N_1(h)$ ↳ numerical method, h discretization parameter / inversely proportional to N or n number of degrees of freedom /---

Examples:

$$N_1(h) = \frac{f(x_0+h)-f(x_0)}{h} \quad \text{or} \quad N_1(h) = \frac{y_{i+1}-y_i}{h} - f(t_i, y_i) \quad \text{etc.}$$

Idea: Combine $N_1(h)$ for different h to improve order of accuracy

$$N_1(h) = M - K_1 h - K_2 h^2 - K_3 h^3 - \dots \quad (A)$$

$$N_1\left(\frac{h}{2}\right) = M - K_1\frac{h}{2} - K_2\frac{h^2}{4} - K_3\frac{h^3}{8} - \dots\dots \quad (B)$$

$$\underbrace{2\,N_1\left(\frac{h}{2}\right) - N_1(h)}_{N_2(h)} = \underbrace{2M - K_1 h - K_2\frac{h^2}{2} - K_3\frac{h^3}{4}}_{2(B)} + O(h^4)$$

$$\underbrace{-M + K_1 h + K_2 h^2 + K_3 h^3 + O(h^4)}_{-(A)}$$

$$= M + K_2\frac{h^2}{2} + \frac{3}{4}K_3 h^3 + O(h^4)$$

$$N_2(h) = M + K_2\frac{h^2}{2} + \frac{3}{4}K_3 h^3 + O(h^4)$$

$$N_1(h) = \frac{f(x_0+h)-f(x_0)}{h} \qquad N_1\left(\frac{h}{2}\right) = \frac{f\left(x_0+\frac{h}{2}\right)-f(x_0)}{\frac{h}{2}}$$

$$N_2(h) = 2\,N_1\left(\frac{h}{2}\right) - N_1(h) = \frac{4\left(f\left(x_0+\frac{h}{2}\right)-f(x_0)\right)}{h} - \frac{f(x_0+h)-f(x_0)}{h}$$

$$= \frac{4T(x_0+\frac{h}{2}) - T(x_0+h) - 3T(x_0)}{h}$$

Now if we have $N_p(h) = M - K_{p,0}\, h^p - K_{p,1}\, h^{p+1} - \dots$ (A)

$$N_p\left(\frac{h}{2}\right) = M - K_{p,0}\left(\frac{h}{2}\right)^p - K_{p,1}\left(\frac{h}{2}\right)^{p+1} - \dots$$

$$= M - \frac{K_{p,0}}{2^p}\, h^p - \frac{K_{p,1}}{2^{p+1}}\, h^{p+1} - \dots \quad (B)$$

2^p (B) $-$ (A) $=$ $\underbrace{2^p M - K_{p,0} h^p - \frac{K_{p,1}}{2} h^{p+1} - \dots}_{2^p \text{ (B)}}$

$$\underbrace{-M + K_{p,0} h^p + K_{p,1} h^{p+1} + \dots}_{-\text{ (A)}}$$

$$= (2^p - 1) M + \frac{K_{p,1}}{2} h^{p+1} + \dots$$]←

$$\Rightarrow \boxed{N_{p+1}(h) = \frac{1}{2^p - 1}\left(2^p N_p\left(\frac{h}{2}\right) - N_p(h)\right)}$$

$N_{p+1}(h) = M + \frac{\quad}{2(2^p-1)} h^{\quad} + \dots$

Romberg Integration

Apply Richardson extrapolation to composite trapezoidal rule

$$T(h) = \int_a^b f(x)dx + \frac{b-a}{12} h^2 f''(\mu) \quad \text{for some } \mu \in (a,b)$$

$$= \underbrace{\int_a^b f(x)dx}_{=M} - K_1 h^2 - K_2 h^4 - K_3 h^6 - \dots$$

we did not prove this, but you can assume it is true

here $T(h) = N_2(h)$ because first error term is $O(h^2)$

Formula from before $N_{p+1}(h) = N_{p+2}(h) = \frac{1}{2^p-1}\left(2^p N_p\left(\frac{h}{2}\right) - N_p(h)\right)$

because we only have even order error terms

Plug in $p=2$: $N_3(h) = N_4(h) = \frac{1}{3}\left(4 N_2\left(\frac{h}{2}\right) - N_2(h)\right)$

$$= \frac{4}{3} T\left(\frac{h}{2}\right) - \frac{1}{3} T(h)$$

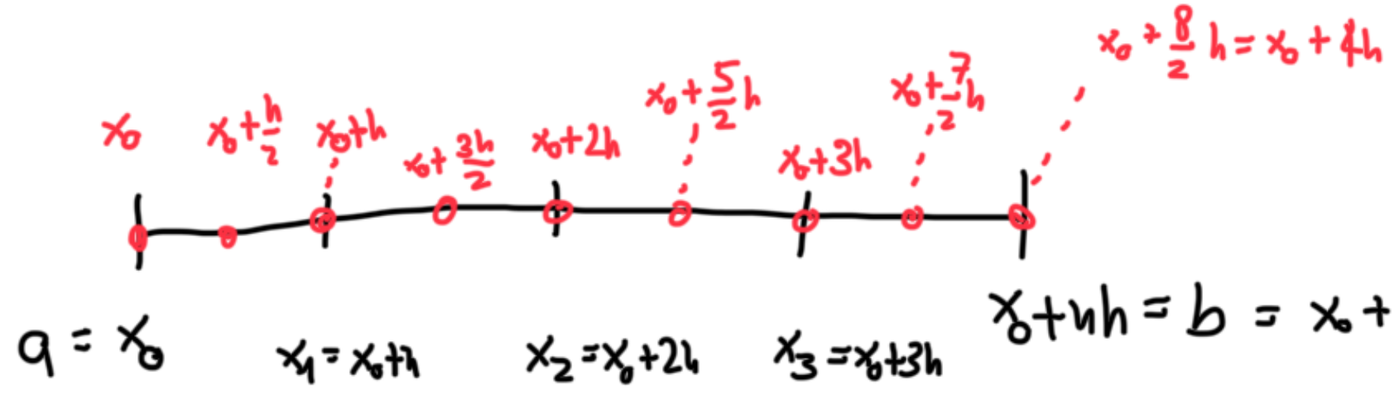
Explicit computation of $N_4(h)$

$$T(h) = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right]$$

$$x_i = x_0 + ih$$

$$= \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_0 + ih) + f(x_0 + nh)\right)$$

$$T\left(\frac{h}{2}\right) = \frac{h}{4}\left[f(x_0) + 2\sum_{i=1}^{2n-1} f\left(x_0 + i\frac{h}{2}\right) + f\left(x_0 + 2n\frac{h}{2}\right)\right]$$



$$N_4(h) = \frac{4}{3} T\left(\frac{h}{2}\right) - \frac{1}{3} T(h)$$

$$= \frac{h}{3}\left[f(x_0) + 2\sum_{i=1}^{2n-1} f\left(x_0 + \frac{ih}{2}\right) + f(x_0+nh)\right]$$

$$- \frac{h}{6}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_0+ih) + f(x_0+nh)\right]$$

$$= \frac{h}{6}\left(f(x_0) + f(x_0+nh)\right) + \frac{h}{3}\left[2\sum_{i=1}^{2n-1} f\left(x_0 + \frac{ih}{2}\right) - \sum_{i=1}^{n-1} f\left(x_0 + 2i\frac{h}{2}\right)\right]$$

$$= \sum_{i=1}^{n-1} f\left(x_0 + 2i\frac{h}{2}\right) + 2\sum_{i=0}^{n-1} f\left(x_0 + (2i+1)\frac{h}{2}\right)$$

$$2\sum_{j=1}^{n-1} f\left(x_0 + 2j\frac{h}{2}\right) + 2\sum_{j=0}^{n-1} f\left(x_0 + (2j+1)\frac{h}{2}\right)$$

$$= \frac{h}{6}\left[\overbrace{f(x_0)}^{a} + \overbrace{f(x_0+nh)}^{b} + 4\sum_{i=0}^{n-1} f\left(x_0 + \frac{(2i+1)h}{2}\right) + 2\sum_{i=1}^{n-1} f(x_0+ih)\right]$$

Midterm 2: $M = N_1(h) + K_1 h + K_3 h^3 + O(h^5)$ h small parameter

(a) Find a, b, k s.t. $N_3(h) = a N_1(h) + b N_1\left(\frac{h}{2}\right)$ satisfies

$$M = N_3(h) + k h^3 + O(h^5)$$

Solution:

$$N_1(h) = M - k_1 h - k_3 h^3 + O(h^5)$$

$$N_1\left(\frac{h}{2}\right) = M - k_1 \frac{h}{2} - k_3 \frac{h^3}{8} + O(h^5)$$

$$2 N_1\left(\frac{h}{2}\right) - N_1(h) = 2M - k_1 h - \frac{k_3 h^3}{4} + O(h^5) - M + k_1 h + k_3 h^3 + O(h^5)$$

$2 \to b$, $-1 \to a$, $*2$

$b = 2 \quad a = -1$

$$= M + k_3 \frac{3}{4} h^3 + O(h^5)$$

$k_3 \frac{3}{4} \to -k$